



Society of Petroleum Engineers

SPE-227451-MS

Porosity Type Quantification and Rock Typing of Ediacaran Complex Carbonate Reservoir (Khufai) Using Advanced Bore Hole Image Analysis

M. Mahruqi, S. Shukairi, and K. AL Rahbi, CC Energy Development, Muscat, Sultanate of Oman; R. Lawati and A. Hamadi, SLB, Muscat, Sultanate of Oman

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227451-MS](https://doi.org/10.2118/227451-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Carbonate reservoirs are famous of being notorious and complex, and challenging for modeling. The Ediacaran Khufai formation is one of the oldest commercially proven petroleum plays in the world located in sultanate of Oman. It Imposes challenges in sub surface characterization due to its post depositional diagenetic and structural overprints. A holistic approach is necessary to capture different scales of the heterogeneities and complement the traditional Petrophysical evaluation in a feedback loop to modeling.

A Methodology developed on two key pillars; The first one lies in identification and quantification of different rock fabrics, which specifies the porosity type: matrix, fractures and vugs that are not easily quantified using conventional open hole logs without a full core. Secondly, linking each of these rock fabrics to a specific petrophysical workflow for the fundamental reservoir petrophysical properties, water saturation and permeability.

The high-resolution borehole images complement the conventional petrophysical evaluation. It provides high vertical and azimuthal resolution Images for geological features like natural fractures and textural information with its spatial arrangement in the borehole wall. The high-resolution images' quantitative heterogeneity mapping and partitioning of matrix and vugs/fractures showed a good match with core analysis allowing an integration of conventional petrophysical logs. It was applied on a data set of number wells resulting in a detailed differentiation of pore types at different scales for rock typing.

The new rock types scheme including the vuggy and fractures were used for petrophysical evaluation. It re-defined reservoir/non-reservoir intervals which impacted the final product of petrophysical model. Furthermore, it supported with surveillance and production logs and facilitates for optimization of well planning and completion.

This workflow fills the gap of the impact of heterogeneities such as m variable due to fractures or vugs and its impact on saturation computation and permeability. It also demonstrates their influence not only the storage but also transmissibility in the form of fractures and or the characters of vugs into modeling workflows and well centric performance. The ultimate goal is to utilize newly developed petrophysical rock types as inputs for static and dynamic models construction pursuing more accurate reserves estimation and a good match with actual well and reservoir performance

Introduction/Background

The Ediacaran-aged Khufai Formation, part of the Nafun Group (ca. 550-635 Ma), is one of the oldest commercially-proven petroleum plays globally (Said Al Balushi 2024). Located in east-central Oman, it features a complex fractured and vuggy carbonate reservoir supported by prolific source rocks (Khufai carbonates and Masirah Bay Formation siliciclastics) and an efficient top seal (Shuram Formation mudstones). Its depositional environment involved highstand progradation of a microbially-influenced carbonate ramp with varied lithofacies, including mudstones, packstones, grainstone shoals, and peritidal carbonates, influenced by tectonics and siliciclastic input. Post-depositional deformation events, including micritization, cementation, dissolution, fracturing, and compaction, have shaped the reservoir. Despite low matrix permeabilities (0.01-20 mD) and porosities (0.1-10%), reservoir quality is enhanced by fracturing and dissolution, creating a dual-porosity system reliant on vug connectivity through fracture networks.

Robust characterization and modeling are essential for exploration and development due to significant spatial variability, rock-fluid interaction and evaluation approach of each reservoir property. That is starting from robust identification and quantification of each pore type and consequently rock types.

Storage wise, primary/matrix porosity is the main storage space as it the most abundance pore type and fundamental unit of the reservoir. Vugs is the biggest pore space as unit and it can be considered as enhanced matrix pore space especially if the dissolution affects the pore throat as well. The third pore space, fractures, has the least storage potential.

In terms of permeability, the behavior is reversed: fractures dominate fluid flow, while the matrix remains predominantly tight, as previously noted. Vugs permeability is subjective to the connectivity to other vugs and fractures. This role exchange and the balance of each existence is the main drive behind naturally fractured carbonate reservoir production profile. If it is dominated by fractured, it produces with high initial rate and drops drastically. When it dominated by matrix porosity, usually it sustains for longer time with lower initial rate.

The other crucial impact on genuine reservoir feature is saturation evaluation. As saturation evaluation approach is Archie, (AGUILERA 2008) has investigated presence of fractures on archie cementation exponent (m). It finds out that including fracture dip and tortuosity in the petrophysical analysis can generate significant changes in the dual porosity exponent (m) of the composite system of matrix and fractures and not taking fracture dip and tortuosity into consideration can lead to significant errors in the calculation of water saturation.

Evaluation challenges

Multiple workflows were explored for conventional log analysis and the approach best suited to log availability and the formation type was selected. Previous workflow that can be implemented using conventional logs only sensitive to lithology (matrix properties) we took a decision to develop a sequential rock typing scheme, whereas first main lithology units are identified using neural network trained on spectroscopy lithology evaluation. The network is using conventional logs only. At a second step we try to identify different rock types for different lithology. Three separate workflows are created for Dolomite, Limestone and Sandstone using conventional logs and NMR data, supervised by NMR FA facies for advanced logs Workflow. Using conventional logs only for conventional workflow, supervised by advanced logs neural network results.

This workflow was excellent on generating mineralogy-based rock types. It even goes extensively by having a total of 14 rock types. Yet, there are two critics on this workflow. First, from engineering point of view, highly detailed mineralogical rock types do not have significant improvement in the modeling. The second drawback of this workflow is not being able to identify the physical attributes of carbonate rocks such as fracture and vugs.

To address these raises, most have validation criteria. For that, production profile log data (SNL) is used. Looking to Figure 2, although Zone A & C has highest porosity and same rock type (Dolomite), Noise log array doesn't indicate that these zones are the best flowing zones. In contrast, although Zone B has lower porosity compared to previous Zones, contribute way better than them.

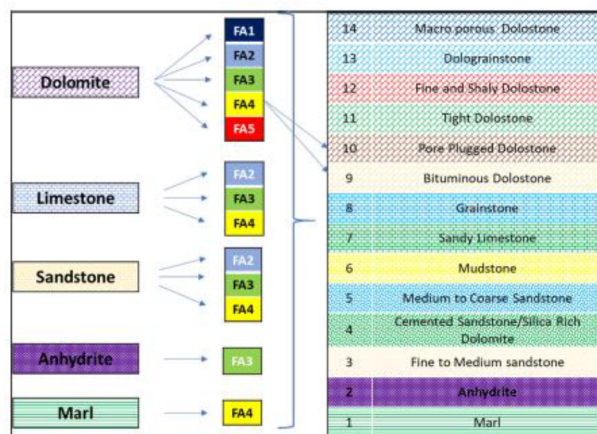


Figure 1—Rock types from previous workflow.

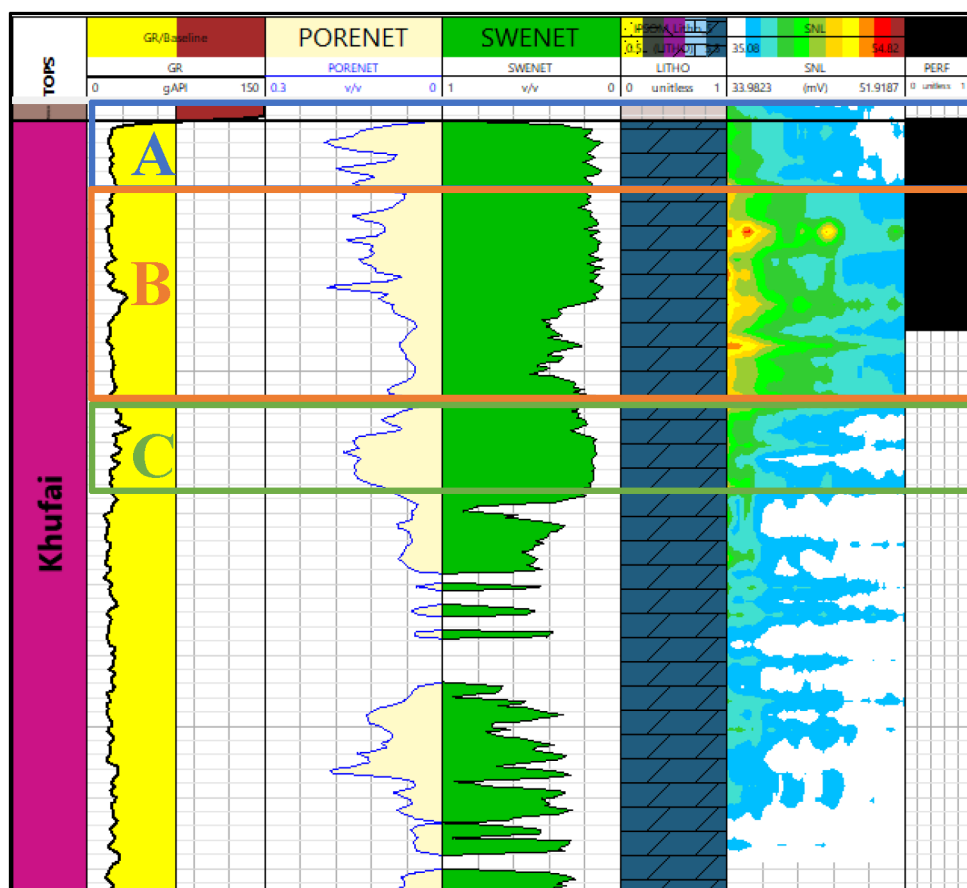


Figure 2—Validate the rock type and porosity evaluation using logs of sonic noise tool array.

Novel approach

The new approach is split into two parts. First, Advanced BHI analysis (Tetsushi Yamada 2013), principal and steps. The proposed workflow starts with preprocessing the raw image data in a series of steps. After

calibrating the image value using an external shallow resistivity log and converting the value to conductivity, the image can be treated as a conductivity map of the borehole wall. The borehole image is displayed as a 2D unrolled map of the borehole viewed from inside. The lateral axis is generally from 0 to 360° from the north in vertical wells. Once the image is preprocessed, bedding planes need to be delineated on the image, so that properties computed on the image can be characterized along the bedding. Solution enhanced bed boundaries are also delineated so heterogeneities can be subclassified using them in heterogeneity delineation. In the following sections, the concepts and the techniques used in the proposed workflow are introduced [Figure 3](#). This paper does not focus on the details of the techniques, but rather on how they can help us interpret borehole images in-term of reservoir properties.

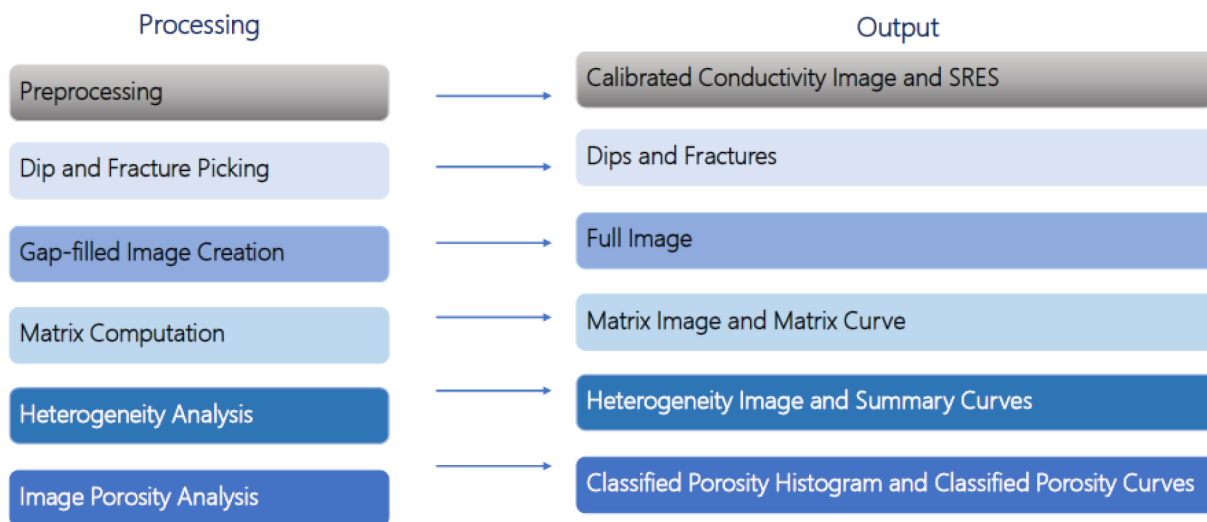


Figure 3—Bore hole image processing steps.

Dip and Fracture picking

When the processing is performed a manual dip picking will be carried out to delineate bed boundaries, dissolutions boundaries, stylolite, fractures, and faults.

Gap-filled image creation. The design of the BHI tool with pads and flaps creating gaps stripes in between those pads & flaps. The larger the borehole diameter, the larger the width of missing strips. For example, the azimuthal coverage is 80% in an 8.5-in borehole. These strips complicate the automatic processing of the image, in particular heterogeneity delineation. The texture analysis provides satisfactory results when the heterogeneity size is smaller than the pad, but becomes inaccurate as soon as the size of the textural features exceeds the pad width. Generating an image in which the missing strips are filled with some appropriate data is a good way to get rid of such a problem and improve final results. Algorithm used for that called multipoint statistics (MPS) ([T. 2006](#)) ([Zhang 2009](#)).

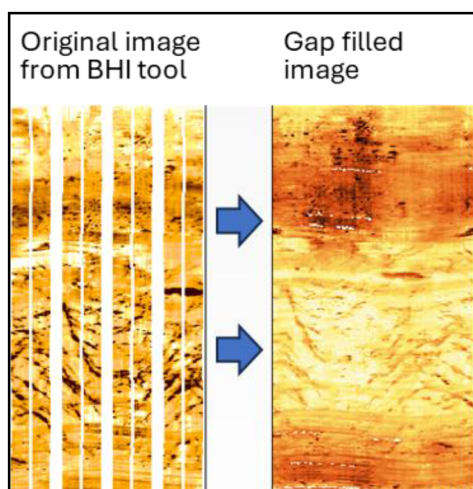


Figure 4—Gap filled image creation.

Matrix extraction. The background of the image, which corresponds to the geological term matrix, is computed by removing noncrossing features on images such as vugs, molds, fracture segments, and intraformational conglomerates. In this way, conductive heterogeneities are firstly removed, then resistive heterogeneities, leaving only the matrix (Figure 5).

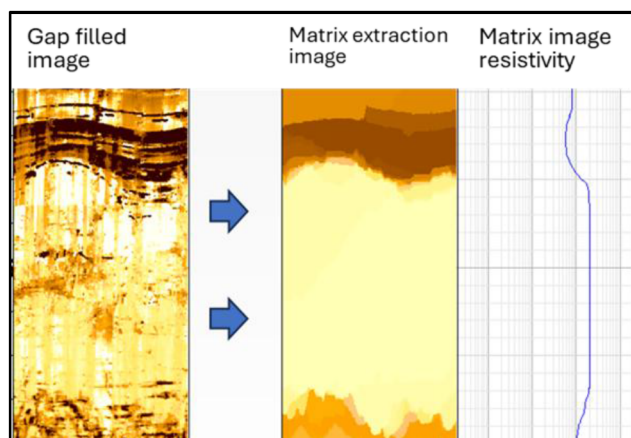


Figure 5—Matrix image extraction.

The property of the matrix image is presented as a curve by averaging the matrix image, respecting bedding layers (Figure 5). This curve is widely used today for variety of purposes such as sequence analysis thanks to its finer resolution and sampling rate compared to conventional resistivity logs.

Heterogeneity Delineation

Before running the delineation of the heterogeneity, the images are broken into small segments by utilizing the matrix images. The output image is referred to as a Mosaic image. Each mosaic piece will have some attributes like conductivity value, contrast value to matrix image, size, and type. By applying cutoff value on the conductivity value and contrast value of mosaic pieces on facies-by-facies basis the mosaic pieces will be delineated into conductive heterogeneity (e. g. vugs (solutions enhanced large or cluster of small vugs or patchy porosities visible to borehole images) or resistive heterogeneity. The conductive heterogeneity reclassified based on the dip picking (beddings & Fractures) into Connected, Isolated, Fractured and at bed boundary conductive heterogeneity.

Image porosity analysis. The porosity map is generated through [equation 1](#) and associated with the heterogeneities computed in the heterogeneity delineation step to classify the porosity values;

$$\phi_i = \phi_{ext} (R_{ext} C_i)^{1/m} \quad (1)$$

Where ϕ_{ext} and R_{ext} are the porosity and the shallow resistivity, respectively, from conventional logs and C_i is a value of the conductivity image. For example, a value in the porosity image which is at a connected conductive spot in the heterogeneity image is classified as porosity from connected conductive spot. As an alternative workflow, heterogeneity delineation can be run on the porosity image to directly delineate pore space. A pore space is a conductive heterogeneity, but conductive heterogeneity is not necessarily a pore space. Thus, the heterogeneity image needs to be refined with the local geological knowledge so that clay and conductive minerals as well as breakouts and induced fractures, which are high conductivity zones but not true pore space, are not considered as high-porosity zones. The histograms of each texture class are created over vertical windows (along the borehole depth) and are stacked in the same track ([Figure 6](#)).

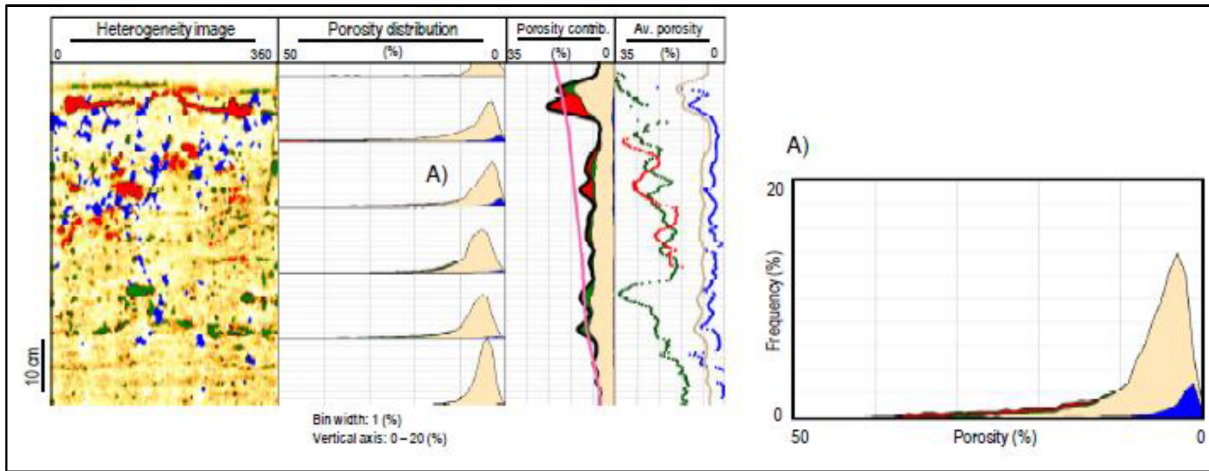


Figure 6—Results of image porosity computation.

The image of total porosity is given by

$$\phi_{ImageTotal} = \frac{1}{I} \sum_i^I \phi_i = \frac{1}{I} \sum_i^I \phi_i = \frac{1}{I} \sum_i^I \phi_i \quad (2)$$

where I correspond to the number of the image column, and t represents each texture class. The image total porosity should follow the global trend of external porosity when the image calibration has been successfully done. Two types of curves are created for each heterogeneity class. The first one is the contribution of each texture category to the total image porosity, which is defined by

$$\phi(t)_{contribution} = \frac{1}{I} \sum_{i \in t} \phi_i \quad (3)$$

The second one is the average porosity of each texture class.

$$\phi(t)_{ave} = \frac{1}{I(t)} \sum_{i \in t} \phi_i \quad (4)$$

where $I(t)$ is the total number of pixels of the texture category.

The second part is generating rock typing based on porosity analysis from Bore hole image. As stated earlier, the final rock typing will be a build upon mineralogical rock typing generated by neural networks. Originally, rock types clay, anhydrite, sandstone, limestone and dolomite were generated. As this was found not sufficient to properly characterize the reservoir and set a solid base to allocate the suitable petrophysical properties and evaluation approaches, the need to consider the nature of rock textural as come. Also, since

the reservoir is mainly dolomitic rock and avoiding over complexity, detailed rock typing covers only the dolomite. Hence, Dolomite rock is subdivided into four textural rock types. First is micro-porous (RT-5) dolomite where only primary porosity existing. Regardless of porosity variation it has, from low as (0.01 v/v) to high as (0.20 v/v),

The vuggy dolomite (RT-6) is the second type of dolomite rock type. As per Lucia porosity classification, vugs are classified into isolated and connected vugs. Where the flow between isolated vugs must pass through matrix porosity and permeability to drain the vugs; therefore, the contribution of separate vugs to total reservoir porosity and permeability can be estimated if matrix characteristics and total porosity are known. And for the connected vugs, the flow is going through easier path. Hence, the permeability and even flow path of resistivity tools current i.e, cementation archie exponent (m) is different. The image analysis algorithm is coded in a way to classify the vuggy porosity into isolated and connected vugs. But, from an image log, that can be only determined in one dimension only. Where the data has to be in three-dimensions to have the feature fully covered the spatial geometry of the rock (something like CT scan 3D models). For this reason, vugs are all merged as one set.

The third type is fractured dolomite (RT-7). First, when mentioned fracture here, they are conductive (from BHI) fractures only. Because we assumed that the conductive fractures are the effective ones. They could be continuous or partial/discontinuous. In contrast, any resistive fractures assumed to be ineffective, hence, not counted as porosity. Second, vugs intersected with fracture are considered part for the fracture. This has to be mentioned because the fracture known to be low porosity volume space, in the workflow you will notice that its porosity value goes high. As the fracture is not a storage space, to classify the rock fractured rock, it only considers the presence of the fractures regardless of the volume it adds to total porosity. And because the classification criteria is based on porosity, the cut off set to >0.0 v/v

When both vugs and fractures co-exist, they just complete each other. High storage capacity rock, if they are tight, they are naturally stimulated by fractures. Vuggy and fractured dolomite is (RT-8).

```

1 mv=MissingValue
2 LOOP:
3   >
4   > if qr != mv and lim !=mv and dol != mv and cly != mv and anh !=mv:
5   >   litho=[qr, lim, dol, cly, anh]
6   >   rt=max(litho)
7   >   #print (rt)
8   >   if rt==cly:
9   >     mrt=1
10  >   elif rt==qr:
11  >     mrt=4
12  >   elif rt==lim:
13  >     mrt=2
14  >   elif rt==dol:
15  >     mrt=5
16  >   elif rt==anh:
17  >     mrt=3
18  >   else:
19  >     mrt=mv
20  >   if porotex!=mv:
21  >     if vugs>VugsCutoff and mrt==5:
22  >       mrt=6 #vug reservoir
23  >     if frac>FracCutoff and mrt==5:
24  >       mrt=7 # fractured reservoir
25  >     if vugs>VugsCutoff and frac>FracCutoff: #and mrt==5:
26  >       mrt=8 # vuggy and fractured
27  >   else:
28  >     mrt=mv
29  >   #print(mrt)
30

```

Figure 7—python script to generate rock type flag using outputs of neural network and porosities from Advanced pore hole images.

Results

Using Bore hole image advanced analysis, porosity spectrum enclosed of matrix pores, isolated and connected vugs, and fractured porosity is generated [Figure 8](#).

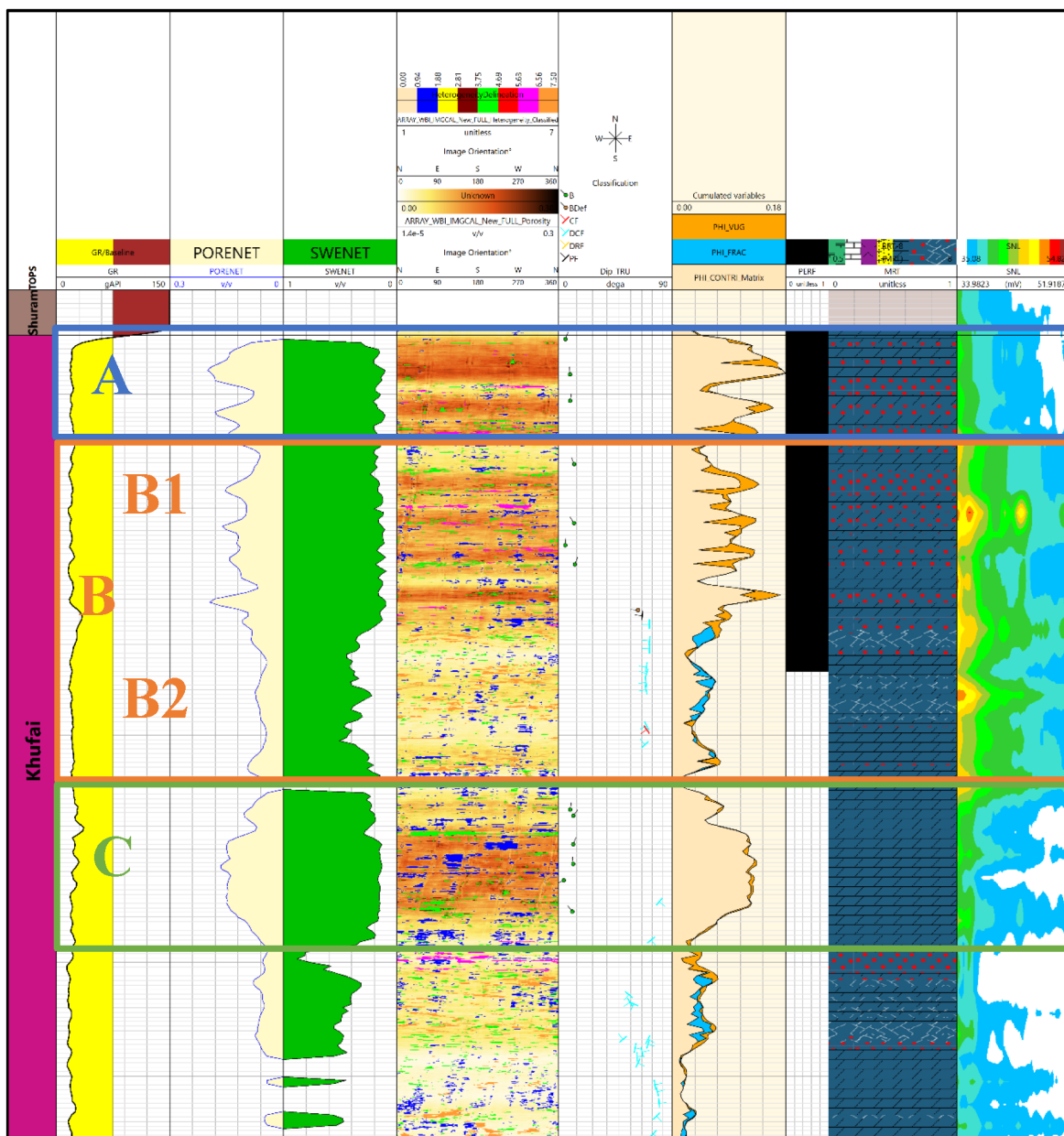


Figure 8—Results of Advanced borehole analysis results and corresponding rock types with sonic noise log.

Enhanced rock types demonstrated in Table 1 are plotted in the well log (Figure 8). The plot shows the total porosity (track #3) full image log with secondary porosities, formation dips and fractures (track #6) final porosity array that includes matrix porosity vugs and fractures porosity (track #7), final rock types (track #9) and finally production profile from array noise tool (track #10). The results are plotted with conventional porosity curve to demonstrate the improvement and production profile track for validation.

Table 1—Final Rock types of resultants of Mineralogical facies and Textural facies.

Mineralogical rock types/Facies	Textural Rock Types/Facies	Criteria	Final rock types	Legend
Clay/Shale	-	max fraction volume	RT-1	
Limestone	-	max fraction volume	RT-2	
Anhydrite	-	max fraction volume	RT-3	
Sandstone	-	max fraction volume	RT-4	
Dolomite	Micro-porous dolomite	max fraction volume	RT-5	
	Vuggy dolomite	RT-5: PORvug>0.02	RT-6	
	Fractured dolomite	RT-5: PORfrac>0.0	RT-7	
	Vuggy and fractured Dolomite	RT-5: PORvugs>0.02 & PORfrac>0.0	RT-8	

Starting from the top, Zone A, is relatively high porosity (0.08-0.20 v/v) dolomitic reservoir with presence of vugs but no fractures. Noise tool doesn't indicate high contribution that would be expected from such high porosity rock. This is revealed to obviously tight matrix plus absence of fractures. It also, indicate that vugs are most likely are isolated.

The common trait of Zone B is it is the main contributor to the production whereas shown by noise log, although it poses lowest average porosity (0.11 v/v zone B1 and 0.07v/v zone B2) between the interest zones discussed in these lines. It will remain a mystery without advanced bore hole image analysis. Zone B1 is a vuggy dolomite. It is different from vuggy reservoir (Zone A) by the fraction of vugs volume to matrix porosity volume is higher. This increases the chance that these vugs to be connected and led to enhanced reservoir connectivity which is reflected in the production profile. Zone B1 is most critical one as it has the lowest porosity and seems to be the highest contributor. Visible fractures reveal image logs are most obvious production attribute. Saying it is critical because it will be very easy to consider as non-reservoir due to low apparent porosity where it is the main contributor.

Zone C completing the picture of reversal understanding of conventional por/permeability relation. Although it has consistently high porosity (0.14v/v) it contributes the less to flow. It barely has any secondary porosity nor vugs neither fracture. Porosity is dominated by tight matrix porosity.

Further validation check was done for these results. It is validated against CT (chromatograph topography) image, and they showed excellent matching. The visible pore spaces (vugs) seen from CT image are captured by image log analysis model. Looking closely to both images, you will notice that they are not identical and that has a logical reason. The image log pictures the outer side of the borehole and in this case, it is 8.5" diameter. Whereas the CT image of the core is picturing the opposite side of the borehole. Add to that, the core size is 6.75" diameter. Which means there is a gap between the two images 1.75" wide. Which is sufficient for minor changes take a place. Nevertheless, the overall pattern is comparable.

Another key information revealed by this workflow for reservoir development and economical aspects. Having quantitative evaluation on different pore types will be reflected first, the reserves and, second, the production profile of the field. (Figure 10) shows the matrix porosity is the most abundant and, also, located at the high side as volume. Fracture, in contrast, has small volume and least frequency. Where the vugs are in between. What makes this system complex is it works the opposite way for flow and permeability. Fractures are the main flow space and matrix is tight. Where, again, vugs are dependent on their connectivity vugs-to-vugs or vugs-to-fractures.

Finally, as any evaluation/measurement method there are uncertainties associated with measurements and evaluation algorithms. First, the uncertainty related to the tools, the highest resolution of bore hole imaging tool in the market (used in this case) is 0.2 in. whereas the size of secondary porosity smaller than this number. Especially for fractures where fracture width of 0.002 "poses a permeability of 1 (Darcy) (Lucia 1995) which is classified as excellent permeability (AHR 2008). Second uncertainty related to measurements is the bore hole coverage by imaging tool. It covers 80% in this case (8.5" hole size) the 20% is predicted by software algorithms. In other words, it is not direct measurement. The uncertainty associated with evaluation workflow is human perception, where fractures are picked manually. That is also applicable to highlighting secondary pores in the image log.

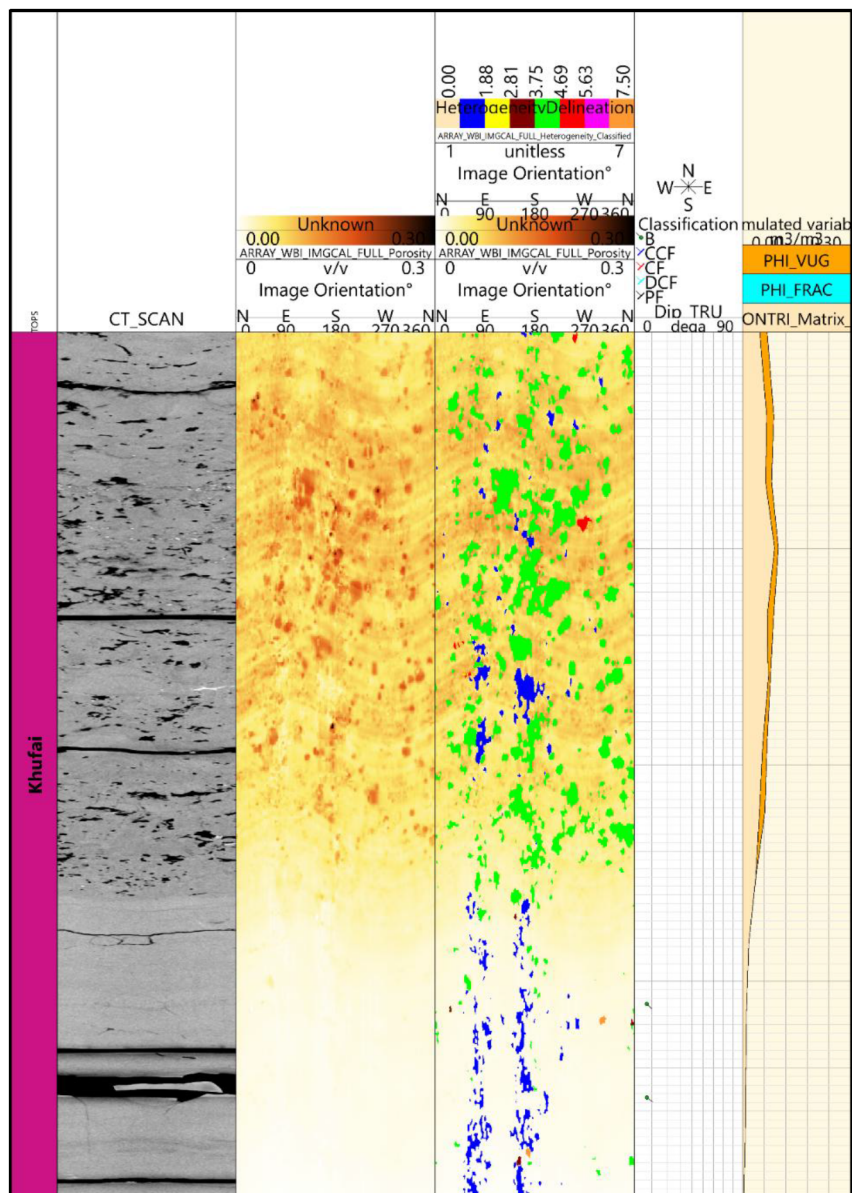


Figure 9—Validate the porosity of different types with CT scan images of core.

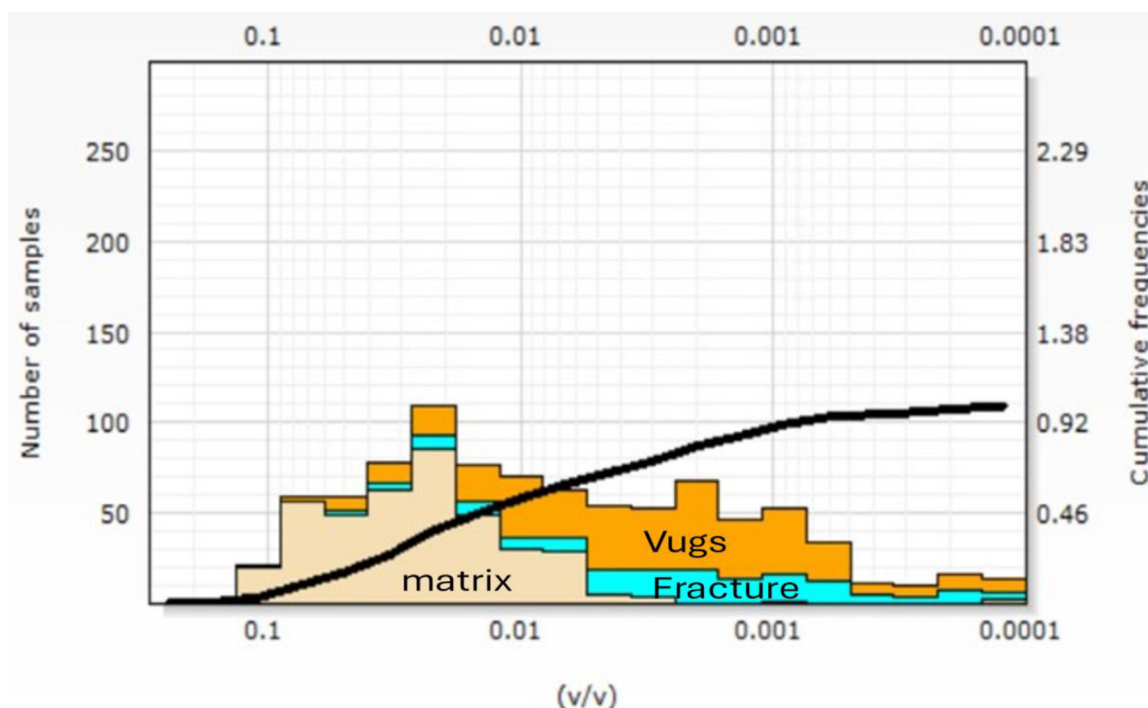


Figure 10—Different porosity types of distribution histogram.

Way forward

Advance bore hole image analysis methodologies, when combined with rock typing methodologies, provide critical inputs for building accurate static reservoir models. This study demonstrates how high-resolution borehole image analysis improves facies definition, dual porosity quantification (matrix, vugs & fractures), and eventually helps in running several volumetric cases—resulting in models that better reflect reservoir reality and its heterogeneity.

Conventional logging often cannot capture these complexities, resulting in oversimplified static models, incorporating advance image analysis and rock typing bridges the gaps which are often not seen in conventional logs. This methodology has assisted in identifying 8 facies within the Khufai Formation, each of the facies has a distinct porosity and permeability trends.

This methodology will improve the accuracy of static and dynamic models by providing detail rock types which includes secondary/diagenetic porosities which are often not captured in the conventional petrophysical models. This will also enhance the development plan, predictability of the complex carbonate reservoirs and reduce the risks.

Conclusion

This study presents an advanced methodology for characterizing complex carbonate reservoirs. The novel approach is to utilize high resolution bore hole image (BHI) analysis combined with conventional petrophysical data. The fabric reservoir features, vugs, fractures significantly influence storage and flow properties. High resolution images enable detailed geological and rock textural characterization, facilitating different pore types and consequently rock types. Integration of these rock type into petrophysical and geological models enhances the accuracy of reservoir delineation, improves the identification of reservoir and non-reservoir zones, and supports better well placement and completion strategies. This integrated approach demonstrates the critical importance of combining advanced imaging techniques with petrophysical analysis for carbonate reservoirs, contributing valuable insights for exploration, development, and production optimization.

Acknowledgment

The authors thank the management of CC Energy Development Oman, Tethys Oil AB, Mitsui Exploration & production Middle East BV and Ministry of Energy and Minerals of Oman for authorizing publication of this work.

References

- Aguilera, R. 2008. "Effect of Fracture Dip and Fracture Tortuosity on Petrophysical Evaluation of Naturally Fractured Reservoirs". Canadian International Petroleum Conference. Calgary, Alberta, Canada: SPE.
- Ahr, Wayne M. 2008. GEOLOGY OF CARBONATE. New Jersey: John Wiley & Sons, Inc.
1995. "Rock-Fabric/petrophysical classification of carbonate pore space for reservoir characterization." In *Carbonate Reservoir Characterization: An Integrated Approach*, by F. Jerry Lucia, 1275–1300.
- Said Al Balushi, Paul Grech, Khalid Al Rahbi, William Fitchen and William Mills. 2024. "Depositional and Diagenetic Characteristics of a Complex Ediacaran Carbonate Reservoir: The Khufai Formation of the East-Central Sultanate of Oman". AAPG. Muscat: ICE.
- T. Zhang. 2006. "Filter-based training image pattern classification for spatial pattern simulation". *PhD dissertation*. Stanford, California, USA.: Stanford University.
- Tetsushi Yamada, Daniel Quesada, Arnaud Etchecopar, Isabelle Le Nir, Jean-Pierre. 2013. "REVISITING POROSITY ANALYSIS FROM ELECTRICAL BOREHOLE IMAGES: INTEGRATION OF ADVANCED TEXTURE AND POROSITY ANALYSIS". SPWLA 54th Annual Logging Symposium. New Orleans, Louisiana: SPWLA.
- Zhang, T., Hurley, N.F., and Zhao, W. 2009. "Numerical modeling of heterogeneous carbonates and multiscale dynamics". SPWLA 50th Annual Logging Symposium.